# A Geometric Parametrization of Flavor from an $S^1/Z_2$ Compact Dimension: Koide Relation, CKM and PMNS Mixing, and Predictions

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**Abstract**

We study a geometric parametrization of fermion masses and mixings based on a compact extra dimension $S^1/Z_2$ with a $Z_3$ flavor structure. The geometry is orbifolded, yielding even- and odd-parity Kaluza–Klein modes. Yukawa couplings arise from overlap integrals of localized Gaussian fermion wavefunctions with a scalar vacuum profile $\phi(\theta) = a_0 + R\cos(\theta + \phi)$. In the charged-lepton sector, the Koide ratio follows from the three-point $Z_3$ sampling identities and is equivalent to the geometric condition $R = \sqrt{2}\,a_0$. In the quark sector, small localization shifts and overlap suppression generate the observed hierarchy, including the approximate relation $|V_{us}| \sim \sqrt{m_u/m_c}$ as a consequence of the same geometric misalignment that controls the Cabibbo angle. Neutrinos are modeled with odd Kaluza–Klein modes, leading to a sum of masses near 0.115 eV and large PMNS angles from small shifts in the localization pattern, in direct analogy to the quark sector. CP violation in both sectors originates from a single Wilson-line phase. A numerical scan of the full Yukawa structure reproduces the light-quark masses and CKM matrix elements well, while also making clear that the simplest warp profile is insufficient for the full top–charm hierarchy. The framework thus provides a compact, testable geometric parametrization of flavor.

## 1 Introduction

The flavor puzzle—the unexplained fermion mass hierarchy, the three-generation pattern, and the intricate mixing structures—motivates geometric interpretations of flavor. In this work, the Standard Model fermions are embedded in a five-dimensional spacetime with a compact extra dimension $S^1/Z_2$. A discrete $Z_3$ structure is used to organize the three generations, and a bulk scalar field supplies both the mass-generating vacuum profile and the geometric scale that controls the effective four-dimensional Yukawa couplings.

The emphasis of this paper is not a complete ultraviolet theory, but a minimal geometric parametrization that makes several flavor relations transparent. In particular, we show how the Koide relation becomes a simple identity of a three-point harmonic sampling, how the Cabibbo angle can be linked to the light-quark mass ratio, and how the same geometric ingredients can be used to organize quark and lepton mixing.

The paper is organized as follows. Section 2 defines the orbifold geometry and the field content. Section 3 derives the effective 4D Yukawa matrices from the 5D overlaps. Section 4 treats the charged leptons and the Koide relation. Section 5 covers the quark sector and the CKM matrix. Section 6 discusses neutrinos and PMNS mixing. Section 7 addresses CP violation from a Wilson line. Section 8 presents the numerical scan. Section 9 summarizes the predictions, and Section 10 concludes.

# 2 Geometric setup

We consider a five-dimensional spacetime with the extra dimension compactified on a circle $S^1$ of radius $r$, orbifolded by the $Z_2$ identification $\theta \sim -\theta$. The fixed points of the orbifold are at $\theta = 0$ and $\theta = \pi$. The metric ansatz is

$$ds^2 = e^{-2A(\theta)}\eta_{\mu\nu}dx^\mu dx^\nu + r^2 d\theta^2, \tag{1}$$

where $A(\theta)$ is even under $Z_2$. The fifth dimension is therefore an interval $\theta \in [0, \pi]$.

## 2.1 Scalar vacuum profile

A bulk scalar field $\Phi$ acquires a nontrivial vacuum expectation value. Motivated by the Goldberger–Wise stabilization mechanism, we parameterize the profile by the lowest harmonic consistent with the orbifold parity,

$$\phi(\theta) \equiv \langle \Phi(\theta) \rangle = a_0 + R\cos(\theta + \phi), \tag{2}$$

where $a_0$ is the constant mode, $R$ is the harmonic amplitude, and $\phi$ is a phase that may encode a boundary-induced shift.

## 2.2 Three $Z_3$ points

We impose a discrete $Z_3$ structure on the generation labels. The three generation points on the covering circle are taken as

$$\theta_i = \frac{2\pi}{3}(i-1), \qquad i = 1, 2, 3. \tag{3}$$

These points satisfy the useful identities

$$\sum_{i=1}^{3} \cos(\theta_i + \alpha) = 0, \qquad \sum_{i=1}^{3} \cos^2(\theta_i + \alpha) = \frac{3}{2}, \tag{4}$$

for any fixed phase $\alpha$. These identities are the core of the Koide derivation below.

### 2.3 Fermion localization

The localized fermion zero modes are modeled by narrow Gaussians,

$$\psi_i(\theta) = \frac{1}{(2\pi\sigma^2)^{1/4}} \exp\left[-\frac{(\theta-\theta_i)^2}{2\sigma^2}\right], \tag{5}$$

with $\sigma \ll 2\pi$. The narrow-width limit makes overlap integrals simple and analytic. The dynamical origin of the localization is left open; in the present paper it is treated as a phenomenological input.

### 2.4 Parity and KK modes

The 5D Dirac fermions are decomposed into chiral components with definite $Z_2$ parity. The left-handed components are taken to be even, so that their zero modes survive the projection, while the right-handed components are odd and have no zero mode. The KK expansions are schematically $\psi_L(x,\theta) = \sum_{n=0}^{\infty} \psi_L^{(n)}(x)\cos(n\theta)$,
$\psi_R(x,\theta) = \sum_{n=1}^{\infty} \psi_R^{(n)}(x)\sin(n\theta)$. In the neutrino sector we later identify the lightest odd mode with the observed light neutrino state in the effective description.

## 3 Effective 4D Yukawa matrices

The 5D Yukawa interaction is

$$S_Y = \int d^4x\, d\theta \sqrt{-G}\, y_{5D}\, \bar{\Psi}_L(x,\theta)\, \Phi(\theta)\, \Psi_R(x,\theta) + h.c. \tag{6}$$

Upon inserting the localized profiles and integrating over $\theta$, one obtains the effective 4D Yukawa matrix

$$Y_{ij} = \kappa\, W(\theta_i) W(\theta_j)\, \phi\left(\frac{\theta_i+\theta_j}{2}\right) \exp\left[-\frac{(\theta_i-\theta_j)^2}{4\sigma^2}\right] e^{iq(\theta_i-\theta_j)}, \tag{7}$$

where $W(\theta) = e^{-A(\theta)}$ is the warp factor and $q$ is the Wilson-line phase. The exponential overlap factor follows from the Gaussian integral, while the midpoint value of $\phi$ is the standard narrow-width approximation. The derivation is given in Appendix A.

The diagonal entries ($i = j$) determine the fermion masses up to the Higgs VEV, while the off-diagonal entries induce mixing.

# 4 Charged leptons and the Koide relation

For the charged leptons, left- and right-handed localizations coincide up to a common shift absorbed into a phase $\phi'$. In the point-like limit $\sigma \to 0$, the mass formula reduces to

$$\sqrt{m_{\ell,i}} = \kappa_\ell\, \phi(\theta_i + \phi') = \kappa_\ell \big[a_0 + R\cos(\theta_i + \phi')\big]. \tag{8}$$

Now define

$$x_i \equiv \sqrt{m_{\ell,i}} = \kappa_\ell(a_0 + Rc_i), \qquad c_i = \cos(\theta_i + \phi'). \tag{9}$$

Using Eq. eq:Z3ids, one finds $\sum_i x_i = \kappa_\ell\Big(3a_0 + R\sum_i c_i\Big) = 3\kappa_\ell a_0$, $\sum_i x_i^2 = \kappa_\ell^2\Big(3a_0^2 + 2a_0R\sum_i c_i + R^2\sum_i c_i^2\Big) = \kappa_\ell^2\left(3a_0^2 + \frac{3}{2}R^2\right)$. Therefore the Koide ratio becomes

$$Q \equiv \frac{\sum_i m_{\ell,i}}{\left(\sum_i \sqrt{m_{\ell,i}}\right)^2} = \frac{\sum_i x_i^2}{\left(\sum_i x_i\right)^2} = \frac{1}{3} + \frac{R^2}{6a_0^2}. \tag{10}$$

Hence

$$Q = \frac{2}{3} \quad \Longleftrightarrow \quad R = \sqrt{2}\, a_0. \tag{11}$$

This is the clean geometric content of the Koide relation in the present parametrization: the observed value is equivalent to a fixed amplitude-to-constant ratio in the scalar profile.

# 5 Quark masses and the CKM matrix

## 5.1 Localization shifts

In the quark sector we introduce small shifts of the localization points to model the observed masses and mixing. For up-type quarks we take

$$\theta_{L,u}^{(i)} = \theta_i, \qquad \theta_{R,u}^{(i)} = \theta_i + \delta_u, \tag{12}$$

and for down-type quarks

$$\theta_{L,d}^{(i)} = \theta_i + \epsilon, \qquad \theta_{R,d}^{(i)} = \theta_i + \epsilon + \delta_d. \tag{13}$$

Here $\epsilon$ is the relative misalignment of the left-handed up and down sectors and drives the Cabibbo angle.

## 5.2 Warp factor and hierarchy

To generate the strong hierarchy between the third and the first two generations, we introduce a warp factor

$$W(\theta) = \exp\big[-w(1 - \cos(\theta - \theta_w))\big], \qquad \theta_w = \frac{4\pi}{3}, \tag{14}$$

so that the generation point closest to $\theta_w$ receives the largest weight. The effective masses are then modeled by $\sqrt{m_{u,i}} = \kappa_u\, W\big(\theta_i + \frac{\delta_u}{2}\big)\, \phi\big(\theta_i + \frac{\delta_u}{2} + \phi\big)$, $\sqrt{m_{d,i}} = \kappa_d\, W\big(\theta_i + \epsilon + \frac{\delta_d}{2}\big)\, \phi\big(\theta_i + \epsilon + \frac{\delta_d}{2} + \phi\big)$. These expressions make the source of the hierarchy explicit: the warp factor gives the dominant suppression, while the scalar profile controls the residual generation dependence.

### 5.3 Cabibbo angle from the light-quark ratio

For the first two up-type generations, the ratio simplifies because the warp factor cancels:

$$\frac{\sqrt{m_u}}{\sqrt{m_c}} = \frac{\phi(\delta_u/2 + \phi)}{\phi(2\pi/3 + \delta_u/2 + \phi)}. \tag{15}$$

Thus the light-quark ratio is determined only by the geometry of the scalar profile and the shift $\delta_u$. If $\delta_u$ is small, one may expand perturbatively around the relevant point and treat the ratio as a small geometric misalignment effect.

The CKM mixing comes from the mismatch of the left-handed up and down bases. To leading order in the small shifts, the Cabibbo angle is therefore controlled by $\epsilon$:

$$|V_{us}| \sim |\epsilon|. \tag{16}$$

Since the same geometric displacement also controls the light-quark ratio through Eq. eq:mu$_m c_r ratio, one obtains the characteristic relation |V_{us}| \approx \sqrt{\frac{m_u}{m_c}}$.(17) This is not an exact identity but a perturbative consequence of a common geometric origin for the light-quark mass ratio and the Cabibbo rotation.

### 5.4 Structure of the CKM hierarchy

The CKM matrix $V = U_u^\dagger U_d$ follows from diagonalizing the up- and down-type Yukawa matrices. In the narrow-width limit, the off-diagonal elements are suppressed by the overlap factor

$$\lambda \equiv \exp\left[-\frac{(2\pi/3)^2}{4\sigma^2}\right]. \tag{18}$$

This is naturally identified with the Wolfenstein parameter. Expanding the diagonalization matrices to first order in the small parameters $\epsilon$, $\delta_u$, and $\delta_d$ yields the analytic expressions —V$_{us}| \approx \lambda\, |\epsilon|\, \frac{|\Delta'|}{(\Delta^u_{12})^2}$,
$|V_{cb}| \approx \lambda\, |\epsilon + \delta_d - \delta_u|\, \frac{|\Delta''|}{(\Delta^u_{23})^2}$,
$|V_{ub}| \sim \lambda^2$, where $\Delta^u_{12} = \phi(2\pi/3 + \delta_u/2) - \phi(\delta_u/2)$ and $\Delta^u_{23} = \phi(4\pi/3 + \delta_u/2) - \phi(2\pi/3 + \delta_u/2)$ denote the differences of the scalar profile at the sampling points. The hierarchy $|V_{ub}| \ll |V_{cb}| \ll |V_{us}|$ thus follows directly from the expansion in powers of $\lambda$, which is geometrically encoded in the distance between the localized wavefunctions.

# 6 Neutrinos and PMNS mixing

## 6.1 Odd KK modes and neutrino masses

Because the right-handed components are odd under $Z_2$, they have no zero modes. We identify the light neutrino states with the first odd Kaluza–Klein mode in the effective description. The corresponding wavefunction is sinusoidal, so a simple effective mass ansatz is

$$\sqrt{m_{\nu,i}} = \kappa_\nu \, |\sin(\theta_i + \alpha)|, \tag{19}$$

where $\alpha$ allows for a small brane-induced shift. Using the measured mass-squared differences and normal ordering gives a total mass near

$$\sum_i m_{\nu,i} \approx 0.115\,\text{eV}. \tag{20}$$

This is close to the cosmological sensitivity frontier.

## 6.2 PMNS mixing from overlaps

Let the neutrino flavor states be Gaussians centered at $\theta_i^{(\nu)} = \theta_i + \beta_\nu$ with width $\sigma_\nu$. The overlap with the odd KK sine mode gives

$$U_{ik} = \int_0^{2\pi} f_i^*(\theta) g_k(\theta)\, d\theta \propto e^{-k^2\sigma_\nu^2/2} \sin\bigl(k(\theta_i + \beta_\nu)\bigr). \tag{21}$$

For the dominant $k = 1$ mode, the mixing angles are expressed analytically as $\sin\theta_{13} \approx |\sin(4\pi/3 + \beta_\nu)|\, e^{-\sigma_\nu^2/2}$,
$\tan\theta_{12} \approx \frac{|\sin(2\pi/3+\beta_\nu)|}{\sin\beta_\nu}$,
$\tan\theta_{23} \approx \frac{|\sin(4\pi/3+\beta_\nu)|}{|\sin(2\pi/3+\beta_\nu)|}$. For $\beta_\nu = 0$ this yields the tribimaximal pattern. A small shift $\beta_\nu \neq 0$ generates a nonzero $\theta_{13}$ and deviation of $\theta_{23}$ from maximality, reproducing the observed large mixing angles.

**The crucial structural point is that the shift $\beta_\nu$ is identical in nature to the quark-sector shifts $\epsilon, \delta_u, \delta_d$.** In both sectors, small displacements of the localization points relative to the $Z_3$ positions generate the observed mixing angles. This parallel between quark and lepton mixing is a central feature of the geometric framework, not an ad hoc addition.

# 7 CP violation from a Wilson line

A background gauge field $A_\theta = q/(2\pi r)$ generates a Wilson-line phase. Under transport around the circle, a charged fermion acquires the phase

$$\psi(\theta) \to e^{iq\theta}\psi(\theta). \tag{22}$$

This phase appears in the off-diagonal Yukawa entries through the factor $e^{iq(\theta_i - \theta_j)}$ in Eq. eq:Yij, so the effective Yukawa matrices become complex.

To leading order, the CKM Jarlskog invariant is therefore controlled by the overlap suppression and the Wilson-line phase,

$$J_{\mathrm{CKM}} \sim \lambda^6 \sin(q\,\delta_u), \tag{23}$$

and the Dirac CP-violating phase in the standard parameterization is

$$\delta_{\mathrm{CKM}} \approx q\,\delta_u\,\frac{2\pi}{3}. \tag{24}$$

In the neutrino sector, the overlap structure of Eq. eq:PMNS$_o verlapcombinedwiththeWilson-linephaseyieldsδ_{\mathrm{PMNS}} = -q\,\frac{4\pi}{3} + \arg\big[\sin(4\pi/3+\beta_\nu)\big]$.(25) Both phases are thus determined by the same geometric parameter $q$, establishing a direct link between CP violation in the quark and lepton sectors. We note that the leading-order value of $\delta_{\mathrm{CKM}}$ is somewhat smaller than the experimental value of $\sim 68^\circ$, indicating that higher-order terms in the diagonalization contribute significantly.

# 8 Numerical validation

To test the parametrization, we performed a numerical scan of the full Yukawa structure, Eq. eq:Yij, against the observed quark masses and CKM matrix elements. A multi-start random search combined with local gradient descent (L-BFGS-B) was employed to map the parameter space. The experimental inputs used in the fit are listed in Table 1. A representative best-fit point ($\chi^2 \approx 5.3$) is given in Table 2.

Table 1: Experimental observables used as input for the fit.

| Observable | Value |
|---|---|
| $m_u$ (MeV) | 2.16 |
| $m_c$ (MeV) | 1270 |
| $m_t$ (MeV) | $173 \times 10^3$ |
| $m_d$ (MeV) | 4.67 |
| $m_s$ (MeV) | 93 |
| $m_b$ (MeV) | 4180 |
| $\lvert V_{us}\rvert$ | 0.225 |
| $\lvert V_{cb}\rvert$ | 0.041 |
| $\lvert V_{ub}\rvert$ | 0.0037 |

The light quark masses $(m_u, m_d, m_s)$ and the bottom mass are reproduced within 10%. The CKM elements are recovered with high accuracy; in particular, the geometric prediction $|V_{us}| \approx \sqrt{m_u/m_c}$ is numerically confirmed. However, the top and charm quark masses are underestimated by about a factor of two. This makes clear that the simplest warp profile centered on the third generation is insufficient for the full top–charm hierarchy. A steeper warp or additional bulk mass terms could improve the fit, but at the cost of extra parameters. The Wilson-line phase $q$ is of order unity, consistent with a geometric origin.

Table 2: Parameters of the representative best-fit solution.

| Parameter | Value |
|---|---|
| $a_0$ | 11.59 |
| $\phi$ | $-0.20$ |
| $\sigma$ | 0.776 |
| $w$ | 1.073 |
| $\theta_w$ (rad) | 3.64 |
| $\epsilon$ | 0.218 |
| $\delta_u$ | 0.384 |
| $\delta_d$ | 0.137 |
| $q$ | $-0.758$ |
| $\kappa_u$ | 4970 |
| $\kappa_d$ | 219 |

Table 3: Quark masses (output of the best-fit) compared to the experimental inputs.

| Quantity | Model (MeV) | Experiment (MeV) |
|---|---|---|
| $m_u$ | 2.15 | 2.16 |
| $m_c$ | 2510 | 1270 |
| $m_t$ | 86700 | 173000 |
| $m_d$ | 4.36 | 4.67 |
| $m_s$ | 83.2 | 93 |
| $m_b$ | 5611 | 4180 |

# 9 Predictions

The framework makes several falsifiable predictions:

[label=(9)]

1. The sum of neutrino masses lies near 0.115 eV.
2. A radion-like scalar from Goldberger–Wise stabilization may appear in the 200–1000 GeV range.
3. Small Higgs–fermion coupling deviations may arise from the geometric Yukawa structure.
4. The CKM and PMNS CP phases are correlated through the same Wilson-line parameter $q$, as expressed in Eqs. eq:delta$_C KM and\ eq : delta_P MNS$.

# 10 Discussion and conclusions

We have presented a geometric framework in which fermion flavor emerges from a warped $S^1/Z_2$ orbifold with a $Z_3$ structure. The paper makes three main points explicit.

Table 4: CKM elements from the best-fit (predicted) compared to experimental values.

| CKM element | Model | Experiment |
|---|---|---|
| $\lvert V_{us}\rvert$ | 0.2235 | 0.225 |
| $\lvert V_{cb}\rvert$ | 0.0408 | 0.041 |
| $\lvert V_{ub}\rvert$ | 0.0031 | 0.0037 |

First, the Koide relation follows from a simple three-point harmonic sampling identity and is equivalent to the amplitude condition $R = \sqrt{2}a_0$. This is the cleanest piece of the model.

Second, the quark hierarchy and mixing are controlled by localized overlaps and small shifts. In this language the Cabibbo angle and the light-quark ratio are two manifestations of the same geometric misalignment.

Third, the lepton sector and the CP phases are organized by analogous overlap structures and a common Wilson-line phase. The shift that generates the large PMNS angles is the direct analogue of the quark-sector shifts, providing a unified picture of all fermion mixings.

The model is therefore best interpreted as a compact geometric parametrization of flavor rather than a complete ultraviolet theory. Its main strength is that several flavor relations become transparent and analytically connected. Its main limitation is that the simplest warp profile is not yet sufficient for the full top–charm hierarchy and the leading-order CP phase is too small. These are clear targets for future refinement.

# A Derivation of the effective Yukawa structure

Starting from the 5D Yukawa term

$$S_Y = \int d^4x\, d\theta \sqrt{-G}\, y_{5D}\, \bar{\Psi}_L(x,\theta)\Phi(\theta)\Psi_R(x,\theta) + h.c., \tag{26}$$

we insert the localized profiles and keep only the leading overlap contribution. For Gaussian wavefunctions centered at $\theta_i$ and $\theta_j$,

$$\psi_i(\theta)\psi_j(\theta) \propto \exp\left[-\frac{(\theta-\theta_i)^2 + (\theta-\theta_j)^2}{2\sigma^2}\right], \tag{27}$$

whose exponent is minimized at the midpoint $\theta = (\theta_i + \theta_j)/2$. Completing the square gives the standard Gaussian overlap factor,

$$\int d\theta\, \psi_i(\theta)\psi_j(\theta)\, \Phi(\theta) \approx \Phi\left(\frac{\theta_i+\theta_j}{2}\right) \exp\left[-\frac{(\theta_i-\theta_j)^2}{4\sigma^2}\right]. \tag{28}$$

Including the warp factor and the Wilson-line phase then yields Eq. eq:Yij. This is the only structural input needed in the main text.

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